

Health Advantages of Swimming in Biophilic System Pools

Evidence-Based Research Report

Executive Summary

Biophilic system swimming pools, also known as natural swimming pools, utilize biological processes rather than chemical disinfection to maintain water quality. These systems offer a sustainable, balanced, and chemical-free alternative to traditional swimming pools, potentially reducing exposure to harmful chemical byproducts while providing mental health benefits through nature connection. However, scientific evidence specifically focused on biophilic pool health advantages remains limited, with most research addressing general swimming benefits or natural water safety concerns. Current evidence suggests potential benefits in reducing chemical exposure and enhancing psychological wellbeing, but pathogen control and water quality management present ongoing challenges that require careful consideration.

Definition & Context

Biophilic System Pools are swimming facilities that integrate natural biological processes to purify and maintain water quality instead of relying primarily on chemical disinfectants like chlorine. These systems use water that is clarified and purified naturally through biological processes, mimicking natural ponds, streams, rivers and freshwater wetlands.

Common Features:

- Biological filtration systems using beneficial bacteria
- Aquatic plants for natural water processing
- Reduced or eliminated chemical disinfectant use
- Integration with natural landscape elements
- Separate zones for swimming and biological treatment

Settings of Use:

- Residential properties seeking chemical-free alternatives
- Hotels and resorts emphasizing eco-friendly amenities
- Therapeutic and wellness centers
- Educational facilities demonstrating sustainable practices

- Community pools in environmentally conscious areas

Benefits / Positive Effects

Benefit	Why It Matters	Supporting Evidence
Reduced Chemical Exposure	Eliminates potential health risks from chlorine and disinfection byproducts	Benefits swimmers by avoiding exposure to potentially harmful substances; Traditional pools can cause breathing problems, asthma, and respiratory illnesses from chlorine byproducts
Improved Skin Health	Natural water systems may be gentler on skin and hair	Swimming in natural bio pools can improve skin health; avoids chlorine-related skin irritation
Enhanced Mental Wellbeing	Connection with natural elements reduces stress and improves mood	Immersing in such environments can boost mood, reduce stress, and improve mental health; Presence of water enhances overall sense of wellbeing
Respiratory Benefits	Absence of chloramine vapors may benefit those with breathing sensitivities	Improved respiratory health; Chloramines cause nasal irritation and breathing problems
Increased Relaxation	Natural aesthetic and reduced chemical odors promote calm	Increased relaxation from swimming in natural bio pools
Environmental Sustainability	Reduced chemical runoff and energy consumption	Supports a healthier, chemical-free ecosystem

Risks / Negative Effects

Risk	Potential Impact (mild/moderate/severe)	Evidence / Source
Pathogen Contamination	Moderate to Severe	Swimming-related illnesses include diarrhea, skin rashes, swimmer's ear, pneumonia from contaminated water; Natural pools require pathogen performance testing and management
Water Quality Variability	Moderate	Natural systems may have inconsistent purification effectiveness; Natural waters are constantly changing and affected by weather, season, and time of day
Maintenance Complexity	Mild to Moderate	Biological systems require specialized knowledge and monitoring; failure can lead to water quality issues
Temperature Fluctuations	Mild	Less climate control than traditional heated pools may limit year-round use
Algae Growth	Mild to Moderate	Harmful algal blooms can occur, especially during hot weather
Wildlife Interactions	Mild to Moderate	Natural systems may attract insects, amphibians, or other wildlife that some users find undesirable

Toxicity & Safety Profile

Human Toxicity Assessment:

- **Short-term exposure:** Generally low toxicity risk from natural biological systems, but potential for acute illness from waterborne pathogens
- **Long-term exposure:** Insufficient long-term studies specific to biophilic pools; reduced chemical exposure likely beneficial

Dose/Exposure Thresholds:

- No established specific thresholds for biophilic pool exposure
- Traditional pools maintain chlorine at 1-4 ppm and bromine at 3-8 ppm - biophilic systems aim to eliminate these chemicals

Health Agency Positions:

- **CDC:** Recognizes swimming health benefits but emphasizes infection control through proper disinfection
- **WHO/EPA/FDA:** No specific guidelines for biophilic swimming systems found in current research

- Regulatory oversight varies by jurisdiction

Vulnerable Groups:

- **Children:** Higher risk due to immature immune systems and increased likelihood of ingesting water
 - **Elderly:** Compromised immune function may increase infection susceptibility
 - **Immunocompromised individuals:** Significantly higher risk from waterborne pathogens
 - **People with respiratory conditions:** May benefit from reduced chemical exposure but face pathogen risks
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Environmental Impact

Positive Environmental Effects:

- Reduced chemical runoff into groundwater and surface water
- Lower energy consumption (no chemical production/transport)
- Support for local ecosystem biodiversity
- Reduced plastic packaging waste from chemical supplies

Potential Environmental Concerns:

- Larger land footprint required for biological treatment areas
- Potential nutrient loading if not properly managed
- May require more water volume for biological zones

Wildlife Impact:

- Generally positive - provides habitat for beneficial organisms
 - May attract wildlife, which can be viewed as positive or negative depending on context
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Who Should / Should Not Be Exposed

Recommended For:

- Individuals sensitive to chlorine or chemical disinfectants
- People with mild respiratory sensitivities (with proper pathogen control)
- Those seeking connection with natural environments
- Environmentally conscious swimmers

- Individuals with chemical-sensitive skin conditions

Should Exercise Caution:

- Immunocompromised individuals
- Infants and very young children
- Elderly with health conditions
- People with open wounds or infections
- Individuals with severe respiratory compromise

Should Avoid:

- Those requiring sterile aquatic environments
 - People with specific pathogen susceptibilities
 - Individuals unable to follow basic water safety practices
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Safer Alternatives or Mitigation Strategies

For Traditional Pool Concerns:

- UV light + minimal chemical treatment systems
- Saltwater chlorination systems
- Ozone treatment combined with reduced chlorine
- Regular shower before swimming to reduce chemical demand

For Biophilic Pool Safety:

- Hybrid systems combining biological and minimal chemical treatment
- UV sterilization as backup pathogen control
- Regular water quality testing for pathogens
- Clear user guidelines and health screening
- Professional system design and maintenance
- Separate zones for different risk tolerance levels

Risk Reduction Practices:

- Avoid swallowing pool water
- Shower before and after swimming

- Stay out of water when ill
- Maintain proper system operation and monitoring

Comparison

Option	Pros	Cons	Best For
Biophilic Pools	No chemical exposure, natural aesthetics, environmental sustainability, potential mental health benefits	Higher pathogen risk, complex maintenance, variable water quality, limited regulatory oversight	Healthy individuals seeking chemical-free swimming, eco-conscious users, therapeutic settings
Traditional Chlorinated Pools	Reliable pathogen control, established safety protocols, consistent water quality, regulatory oversight	Chemical exposure risks, respiratory irritation, environmental impact, artificial environment	Public facilities, competitive swimming, high-use scenarios, immunocompromised users
Saltwater Chlorination	Reduced chlorine handling, gentler on skin/hair, still provides disinfection	Still produces chlorine, equipment costs, potential corrosion issues	Balance between natural feel and pathogen control
UV + Minimal Chemical	Reduced chemical use, effective pathogen control, lower environmental impact	Equipment costs, backup chemical systems still needed, technical complexity	Users wanting reduced but not eliminated chemical exposure

Sources

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2. [Natural Pool Designs: Eco-Friendly, Biophilic, Organic Shapes - Coastal Custom Pool](#)
3. [Biodynamic natural swimming pool - Sunrose 7](#)
4. [Six Mental Health Benefits of Swimming - Northern Healthcare](#)
5. [Beach and Swimming Pool Safety: Environmental Toxins and Chemical Concerns - WebMD](#)
6. [Swimming pools and health risks: A parent's evidence-based guide - Parenting Science](#)
7. [Guidelines for Keeping Your Pool Safe and Healthy - CDC](#)
8. [Swimming and Your Health - CDC](#)
9. [Preventing Swimming-related Illnesses - CDC](#)

10. [Natural Pools: Safe for Swimming? - Water Shapes](#)
 11. [Swimming safety: Understanding the risks of natural waters vs pools - Mount Nittany Health](#)
 12. [Swimming Pool Safety - American Chemistry Council](#)
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Research Limitations: Current scientific literature specifically focused on biophilic swimming pool health effects is limited. Most evidence is derived from studies on traditional pool chemical exposure risks, general swimming benefits, and natural water safety considerations. More peer-reviewed research is needed to establish definitive health advantages and safety protocols for biophilic swimming systems.